



Dr. François Gardavaud

MEDICAL PHYSICIST EXPERT (MPE) IN MEDICAL IMAGING | DATA SCIENTIST ADVISOR

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I have been practicing in medical imaging since 2011. I'm specialized in CT modality. I have a deep interest in programming and data science. This resume was even generated with the language.

Professional Experience

Data scientist advisor in medical physics

APHP.SORBONNE UNIVERSITY.

Paris

since 2025

- Data science project, clinical data management, clinical statistics, scientific report, publication assistance.

Medical physicist in medical imaging

TENON UNIVERSITY HOSPITAL - APHP.

Paris

Since 2014

- Patient radiation protection, protocol optimization for CT and interventional room modalities, patient skin doses evaluation in interventional radiology, women foetus doses assessment, deployment and use of 2 dose monitoring systems.
- Exploratory data analysis with R language, signal processing with Matlab language, publication assistance for radiologists.
- IT support: management of post-processing consoles, implementation of DICOM nodes, automation scripts design for Mac stations, R scripts development for extraction and analysis of the department clinical activity ...
- Regulatory quality assurance. QC implementation in MRI.
- Technical support in imaging for my fellow physicists in radiation therapy.

Medical physicist in medical imaging

MONDOR UNIVERSITY HOSPITAL - APHP.

Paris

2011 - 2014

- Patient radiation protection, protocol optimization for X-Ray modalities, patient skin doses evaluation in interventional radiology, women foetus doses assessment, deployment and use of 1 dose monitoring systems.
- Regulatory quality assurance implementation.

Selected Publications

For a complete list of my publications see my Google Scholar profile .

- C. Evin, L. Razakamanantsoa, **F. Gardavaud** et al. *Clinical, Dosimetric and Radiomic Features Predictive of Lung Toxicity After (Chemo)Radiotherapy. Clinical Lung Cancer*. 2025.
- M. Barral, L. Lassalle, **F. Gardavaud** et al.. *Virtual Injection Software Reduces Radiation Exposure and Procedural Time of Prostatic Artery Embolization Performed with Cone-Beam CT J Vasc Interv Radiol*. 2023.
- J. Greffier, Y. Barbotteau and **F. Gardavaud**. *iQMetrix-CT: New software for task-based image quality assessment of phantom CT images. Diagnostic and Interventional Imaging*. 2022.
- F.H Cornelis, L. Razakamanantsoa, M. Ben Ammar, M. Najdaw, S. El-Mouhadi, **F. Gardavaud**, M. Barral. *Percutaneous screw fixation of pelvic bone metastases using cone-beam computed tomography navigation. Diagnostic and Interventional Imaging*. 2022.
- F.H Cornelis, L. Razakamanantsoa, M. Ben Ammar, M. Najdaw, **F. Gardavaud**, S. El-Mouhadi, M. Barral. *Percutaneous Image-Guided Vertebral Fixation in Cancer-Related Vertebral Compression Fractures: A Case Series Study. Medicina*. 2021.
- M. Barral, **F. Gardavaud**, L. Lassalle et al. *Limiting radiation exposure during prostatic arteries embolization: influence of patient characteristics, anatomical conditions and technical factors. European radiology*. 2021.
- M. Savanovic, **F. Gardavaud**, D. Jaros et al. *Contribution of Imaging to Organs at Risk Dose during Lung Stereotactic Body Radiation. J Biomed Phys Eng*. 2021.
- E. Kermarrec, **F. Gardavaud**, K. Kerrou et al. *Risk and safety of breast imaging during pregnancy and lactation. Imagerie de la Femme*. 2020.
- **F. Gardavaud**, S. Tavoraro, N. Grussenmeyer-Mary et al. *Peak Skin Dose evaluation for vascular clinical procedures in interventional radiology: a comparison between three computation numerical. Physica Medica* 2018.
- H. Pasquier, **F. Gardavaud**, M. Chiaradia et al. *Iterative reconstructions in multiphasic CT imaging of the liver: qualitative and task-based analyses of image. Clinical Radiology*. 2018.
- E. Herin, **F. Gardavaud**, M. Chiaradia et al. *Use of Model Based Iterative Reconstruction (MBIR) in reduced-dose CT scan for routine follow up for patients with lymphoma: dose savings, image quality and phantom study. European radiology*. 2015.
- E. Meyblum, **F. Gardavaud**, T-H. Dao et al. *Breast tomosynthesis: Dosimetry and image quality assessment on phantom. Diagnostic and interventional Imaging*. 2015.
- **F. Gardavaud**, A. Luciani, A. Rahmouni. *Radiation exposure from CT scans in childhood and subsequent risk of leukaemia and brain tumours: a retrospective cohort study. Correspondence. The Lancet*. 2012

IT DEVELOPMENT

F. Gardavaud, H. Pasquier, A. Luciani, A. Rahmouni. ProtoEnhance : software to help scan acquisition parameters optimization for the Legacy platform of GE HealthCare Computed Tomography 

F. Gardavaud. Code  to generate interventional radiology DRL and view key patient exposure statistics. 

F. Gardavaud. Code  to generate CT DRL and view key patient exposure statistics. 

PATENT

H. Pasquier, D. Crotty, S. Jacquot Ingles, A. Luciani, **F. Gardavaud**. Methods and apparatus to correct the measurement of water equivalent diameter in computed tomography when patients are miscentered. US patent US10966671B2. Avril 2021. 

Mentoring Experience

Oncological and functional outcomes of image-guided high-dose-rate uterovaginal brachytherapy with single implantation: a French multicenter study.

MASTER 2 STUDENT INTERNSHIP SUPERVISOR.

Sorbonne University, January - August 2025

Radiomic study to identify predictive biomarkers of radiochemotherapy treatment outcome in non-small cell lung cancer

MASTER 2 STUDENT INTERNSHIP SUPERVISOR.

Tours University, February - August 2022

Contribution of metabolic PET-CT imaging combined with radiomics analyzed surveillance imaging in locally advanced non-small cell lung cancer treated with radiochemotherapy to predict survival outcomes and toxicity

MASTER 2 STUDENT INTERNSHIP SUPERVISOR.

Grenoble Alpes University, February - August 2022

Medical physics internship in medical imaging

STUDENTS INTERNSHIP SUPERVISOR.

Tenon University Hospital 2014 - 2020

Validation of an automated solution for organ dose estimation in CT

MASTER 2 STUDENT INTERNSHIP SUPERVISOR.

Clermont-Ferrand University, April - October 2019

Development of an automatic CT image analysis system

MASTER 2 STUDENT INTERNSHIP SUPERVISOR.

Clermont-Ferrand University, February - August 2018

Development of a system to assist in CT protocols personalized optimization

PHD STUDENT SUPERVISOR.

Paris-Sud University, 2014 - 2017

IGRT implementation on a Novalis TrueBeam Stx (Varian)

MASTER 2 STUDENT INTERNSHIP SUPERVISOR.

Clermont-Ferrand University, March - August 2016

Development of a module helping in CT acquisition parameters optimization

MASTER 2 STUDENT INTERNSHIP SUPERVISOR.

Clermont-Ferrand University, March - August 2013

Development of an evaluation module for CT-MRI fusion in brain imaging

MASTER 1 STUDENT INTERNSHIP SUPERVISOR.

Clermont-Ferrand University, May - September 2012

Study and implementation of reference doses levels delivered in interventional cardiology for diagnostic and therapeutic purposes

MASTER 2 STUDENT INTERNSHIP SUPERVISOR.

Paris-Sud University, April - September 2012

Quality controls implementation in CT scanning

BACHELOR STUDENT INTERNSHIP SUPERVISOR.

Créteil University, May - June 2011

Courses were given in French.

Welcoming a new scanner: from acceptance to protocol optimization

CLASSROOM GE HEALTHCARE : EXPERT TO EXPERT PHYSICIENS .

- Program creation, practical work design
- Lecture : Fundamental properties of image quality.
- Lecture : State of the art on CT acceptance
- Lecture: Tips & tricks in CT.

Buc

Since 2019

Image quality principle and CT application specificities

TEACHER IN MASTER 2 SIM

- Course exposing the state of the art of the image quality qualification and quantification with medical physics tools.

Créteil University

2014 - 2022

AI in radiology

TEACHER IN SECOND YEAR UNDERGRADUATE.

- Lecture popularizing artificial intelligence concepts in medicine by illustrating with a medical physics project.

*Faculty of Medicine Sorbonne
University*

2019 - 2021

Advanced CT optimization with iterative reconstructions

POSTGRADUATE EDUCATION .

- Scientific council member : Program creation, practical work design, administrative tasks.
- Lecture: Classical and advanced metrics for CT image quality assessment.
- Lecture : New metrics: can they be used on patients?

Nîmes

2021-10

Patient radiation protection for medical physicists specialized in radiodiagnostics

POSTGRADUATE EDUCATION

- Scientific council member : Program creation, practical work design, administrative tasks.
- Lecture : Image quality metrics.
- Lecture : image quality metrics tailored to CT iterative reconstruction.

Online

2021-01

CT performance

ENSEIGNANT À LA FORMATION DQPRM.

- Course exposing the state of the art of the image quality qualification and quantification with medical physics tools.

INSTN, Saclay

2018 - 2019

Patient radiation protection for medical physicists specialized in nuclear medicine

POSTGRADUATE EDUCATION

- Lecture : Classical and advanced metrics for CT image quality assessment.

Paris

2018-03

Patient radiation protection for medical physicists specialized in radiodiagnostics

POSTGRADUATE EDUCATION

- Scientific council member : Program creation, practical work design, administrative tasks.
- Lecture : Image quality metrics.
- Lecture : image quality metrics tailored to CT iterative reconstruction

Lyon

2017-09

POST-GRADUATE TRAINING

All training courses have been passed in French, except those outside France

06/2024	Data Science essentials with R. 🏆 Certificate of achievement.	PLB Levallois-Perret, France
2023-10	Refresher training in biostatistics with JASP and JAMOVI software.	URFIST Bordeaux, France
2021-12	MOOC : Learn about Data Science and its challenges. 🏆 Certificate of achievement.	France Université Numérique. CY Cergy Paris University
2021-04	MOOC : Multidimensional data analysis. 🏆 Certificate of achievement.	France Université Numérique. Agrocampus. Rennes
2020-11	MOOC : Research Ethics. 🏆 Certificate of achievement.	France Université Numérique. Lyon University.
2019-10	MOOC : Introduction to Statistics with R. 🏆 Certificate of achievement.	France Université Numérique. Paris Sud University.
2019-06	Introduction to clinical data management and analysis with R. Intermediate level.	Cancéropôle Ile-de-France. Paris
2018-12	Postgraduate education : MRI: methodological bases and applications in imaging and radiotherapy.	Clermont-Ferrand
2018-12	MOOC : Introduction to clinical data management and analysis with R. Beginner level.	Cancéropôle Ile-de-France. Paris
2018-12	MOOC : Reproducible Research: Methodological Principles for Transparent Science. 🏆 Certificate of achievement.	France Université Numérique. INRIA.
2018-01	Computed Tomography – Technology, Dosimetry, Optimization. 🏆 Distinction Award.	EFOMP School for MPE, Prague, Czech Republic
2017-10	GE training in MRI sequence programming (EPIC and ORCHESTRA).	GE Training Center, Buc
2017-07	Magnetic Resonance Imaging: Advanced Clinical Applications Safety Aspects, Quality Control. 🏆 Distinction award.	EFOMP School for MPE, Prague, Czech Republic
2017-02	Leadership in Medical Physics: Development of the profession and the challenges for the Medical Physics Expert in Diagnostic and Interventional Radiology. 🏆 Merit award.	EUTEMPE-RX, Module MPE01, Prague, Czech Republic
2016-03	Role of the medical physicist in CT imaging and patient dose optimization. 🏆 Distinction award.	EUTEMPE-RX, Module MPE08, Lausanne, Switzerland
2015-10	Postgraduate education: Image Processing in Medical Physics.	Nantes

UNIVERSITY EDUCATION

October 2025	PhD : Data science and modeling for image quality assessment of thoracic computed tomo- graphy examinations.	ISCD - ED SMAER. Sorbonne University. Paris
2009 -2010	French internship in medical physics.	Institut Curie et Hôpital Militaire du Val-de-Grâce, Paris
2009	French medical physics internship access context. 🏆 Ranked 13th out of 78.	Paris
2008 - 2009	Master 2 - Research, mention Medical Physics. 🏆 Honors.	Paris XI university.

TECHNICAL SKILLS

Main Equipments

GE CT scanners
GE and Hologic
mammographs
GE interventional room
GE and Siemens MRI

Professional software

ADW
ArtiScan

DoseWatch
RDM
ImageJ

MRQuantif
Osirix
Horos

QC material

RTI Piranha
PTW pencil ionization
chamber
Gafchromic XRCT2 films

Phantoms

CATPHAN 600
Kyoto Kagaku PBU-60

Kyoto Kagaku Lungman
Mercury Phantom

IT SKILLS

Programming languages

AppleScript – Git – M+
Markdown – Matlab – R – Shell

Software

FileZilla – Rstudio – Visual Code
Studio

OS

macOS – Linux(Ubuntu) –
Windows

EVENT ORGANIZATION

10/2021	Postgraduate education : Advanced optimization in computed tomography with iterative reconstructions 🔗 .	Nîmes
01/2021	Postgraduate education : Patient radiation protection for medical physicists specialized in radiodiagnostics 🔗 .	En ligne
09/2017	Postgraduate education : Patient radiation protection for medical physicists specialized in radiodiagnostics.	Lyon

LANGUAGES

Compétence	French	English	Spanish
Reading	mother tongue	C2	A1
Writing	mother tongue	C2	A1
Speaking	mother tongue	C1	A1

Common European Framework of Reference for Languages: A1/A2: Basic User. B1/B2: Independent User. C1/C2: Proficient User

Inter-professional activities

2017 - 2023	Vice-president of the medical physics college.	APHP
2017 - 2018	Secretary of the Imaging Section.	French national society of medical physics
2015 - 2016	Medical Physics Expert for Computed Tomography Tenders	AGEPS - APHP

Awards & Distinctions

2017	Medical Physicist Expert (MPE)	EFOMP
2016	Radiation Protection Pavilion Prize Winner	CIRSE, Barcelona
2009	Fellowship	French national society of radiation protection

References

- **Jean-Noel Foulquier, PhD.** 📍 CHU Tenon, APHP, Paris. @ jean-noel.foulquier@aphp.fr
- **Pr. François Cornelis, MD, PhD.** 📍 Memorial Sloan-Kettering Cancer Center, New York, USA. @ cornelif@mskcc.org
- **Pr. Alain Luciani, MD, PhD.** 📍 CHU Henri Mondor, APHP, Créteil. @ alain.luciani@aphp.fr
- **Dr. Leo Razakamanantsoa, MD, MSc.** 📍 CHU Tenon, APHP, Paris. @ leo.razakamanantsoa@aphp.fr

Interests and sport practice

- **Sport** : Bicycle 🚲, fitness session "Les Mills" 🏋️ and running 🏃.
- **Voyage** : Many trips ✈️ to Europe area to discover other cultures; keen interest in Japanese culture.
- **Musique** : Guitar practice 🎸